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An analytical model of a silicon MEMS vaporizing liquid microthruster and some experimental studies

D.K. Maurya *, S. Das, S.K. Lahiri

Microelectronics Laboratory, Advanced Technology Centre/E&ECE Department, Indian Institute of Technology, Kharagpur 721302, India

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Abstract

A recent application of the MEMS technology is in the field of microthrusters for micro/nano satellites. A silicon MEMS vaporizing liquid microthruster (VLM) produces continuously variable thrust in the range from μN to mN. The theoretical simulation of a VLM involves complex numerical 3D micro-fluidic, thermodynamic and electro-thermal solutions. A fast analytical method is, however, desirable in the initial phase of development of a VLM. In this paper, a simple analytical model of a VLM is presented. The model is based on one-dimensional approximations for fluid-dynamical and heat-flow equations. VLMs are fabricated by bonding two micromachined silicon chips. The device consists of a microcavity, an inlet nozzle, an exit nozzle, a microchannel and an internal p-diffused resistor for heating. The thrust is measured by a sensitive cantilever and a laser based lamp-and-scale arrangement. The experimental results on the variation of thrust with heater power are interpreted with the help of the theoretical model. A novel iterative computation is performed to extract the chamber temperature and other important parameters corresponding to the measured values of thrust for different values of heater power. The model gives some physical insight into the operation of the VLM.

Keywords: Microthruster; Micro/nano satellites; Diffused heater; Micromachining; Nozzle

1. Introduction

Micro/nano satellites are drawing considerable attention of space technologists in recent years. This new range of satellites requires micro-propulsion units [1] with an extremely high precision of control. For miniaturizing propulsion systems, the conventional fabrication technology can hardly be used to scale down the size below several inches. The microfabrication technology of MEMS has been successfully employed to batch-fabricate micro-propulsion systems or microthrusters with the dimension in the range of millimetres to sub millimetres, capable of producing extremely small thrusts from micro-Newtons to milli-Newtons.

Of the various types of MEMS microthrusters developed so far, the vaporizing liquid microthruster (VLM) is simple and has been widely investigated [2–4]. VLM can produce continuously variable microthrusts using a non-toxic liquid propellant. The work on VLM reported so far is mostly experimental with little theoretical content [2–4]. The structure of a VLM realised by silicon bulk micromachining is rather complex from the view point of theoretical simulation. It involves rigorous numerical simulations using 3D microfluidic, thermodynamic and electrothermal solvers. No detailed simulation results have been published till date. The 3D numerical simulations are computationally cumbersome and not convenient in the initial phase of development. In an attempt to develop silicon VLM independently, the authors have developed a simple analytical model of VLM for a rough estimation of the parameters involved and for gaining a physical insight into the operation of the VLM. The model is presented in this paper. The paper also includes a brief description of the fabrication and testing of a VLM. While interpreting the experimental results with the help of the theoretical model, the physical principles on which the model is based are established. The analytical model is described

* Corresponding author. Tel.: 91 3222281479; fax: 91 3222282013.
E-mail address: dkm@ece.iitkgp.ernet.in (D.K. Maurya).

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in Section 3 following the presentation of the device configuration in Section 2. The fabrication of the experimental device and the testing and measurement are reported in Section 4. The results are discussed and interpreted theoretically in Section 5. The concluding remarks are given in Section 6.

2. Device configuration for the model

MEMS VLMs developed by various groups [2–4,7] have slightly different configurations, such as (a) propellant inlet from side/top/bottom surface, (b) polysilicon/implanted/diffused heater on one outer surface or on both outer surfaces of the cavity or vaporizing chamber, (c) pyramidally shaped exit nozzles with slanted walls along (1 1 1) planes etched by wet anisotropic etching, steeper pyramidal or cylindrical (de Laval) [3,5] nozzles formed by DRIE, etc. In the present work, an internal p-diffused heater on the bottom chip is used as the microheater [7]. The propellant enters the chamber through a lateral V-groove channel via an inlet nozzle on the top chip. In the present study deionised water was used as the liquid propellant for experimental simplicity. The exit nozzle is pyramidally shaped with (1 1 1) sidewalls formed by wet anisotropic etching. The nozzle is symmetric about the mid plane. The dimensions of the different parts are governed by the mask design, wafer thickness and depth of anisotropic etching. The resistor layout and resistance value are dependent on the diffusion mask and doping parameters. The above geometrical and physical parameters are adjustable.

3. The analytical model

3.1. Thrust equation

As the liquid propellant (deionised water in the present study) enters the vaporizing chamber or cavity through the microchannel at a controlled rate, it is heated up by the inbuilt microheater while moving towards the exit nozzle. The liquid is partially or fully vaporized depending on the applied power to the heater. The vapor emerges through the exit nozzle which is convergent/divergent type. The rate of change of momentum associated with the exit velocity of the vapor mass and the differential pressure at the exit point with respect to the surrounding will give rise to the thrust opposite to the direction of the exit velocity. The thrust is given by [5]:

$$F = \overset{*}{m}\, V_{\text{exit}} + (P_{\text{exit}} - P_{\text{atm}})A_{\text{exit}} \tag{1}$$

where F is the thrust force, m^* mass flow rate of the propellant, V_{exit} the exit gas velocity, P_{exit} the exit pressure, P_{atm} the atmospheric/surrounding pressure and A_{exit} is the area of the nozzle at the exit side.

This is a simple equation based on one-dimensional gas flow approximation. The exit velocity of vapor from the nozzle has been expressed as [5,6]:

$$V_{\text{exit}} = \sqrt{\frac{2\gamma R' T_{\text{c}}}{\gamma - 1}\left\{1 - \left(\frac{P_{\text{exit}}}{P_{\text{c}}}\right)^{\frac{\gamma-1}{\gamma}}\right\}} \tag{2}$$

where T_{c} is the chamber temperature, R' the propellant gas constant, γ the specific heat ratio and P_{c} is the chamber pressure.

For a convergent/divergent nozzle, the pressure ratio $\left(\frac{P_{\text{exit}}}{P_{\text{c}}}\right)$ and the nozzle expansion ratio $\left(\frac{A_{\text{exit}}}{A_{\text{t}}}\right)$ are found to be related as [5,6]:

$$\frac{A_{\text{exit}}}{A_{\text{t}}} = \left[\left(\frac{\gamma+1}{2}\right)^{\frac{1}{\gamma-1}}\left(\frac{P_{\text{exit}}}{P_{\text{c}}}\right)^{\frac{1}{\gamma}} \times \sqrt{\frac{\gamma+1}{\gamma-1}\left\{1 - \left(\frac{P_{\text{exit}}}{P_{\text{c}}}\right)^{\frac{\gamma-1}{\gamma}}\right\}}\right]^{-1} \tag{3}$$

where A_{t} is the throat area of the exit nozzle. It is based on one-dimensional fluid flow through the nozzle with cross-section gradually expanding from A_{t} at the throat to A_{exit} at the exit end. It means that the contour of vapor flow matches with that of the nozzle.

It may be noted that the thrust may be computed, if the values of the chamber temperature and pressure (T_{c}, P_{c}) and the exit pressure (P_{exit}) can be estimated. T_{c} and P_{c} are interrelated thermodynamically. A simple method of estimation of T_{c} and P_{c} as a function of heater power will be discussed later. Hence P_{exit} may be found from Eq. (3) for the given nozzle geometry (A_{t}, A_{exit}).

For the present VLM structure, however, the nozzle exit area, A_{exit}, in Eq. (1), is not the area of the mouth of the exit nozzle formed by the wet anisotropic etching of silicon. The SEM photograph of the exit nozzle of 30 μm × 30 μm throat size is shown in Fig. 1. The Miller indices of the silicon surface are (1 0 0), while the four anisotropically etched side walls are along (1 1 1) planes inclined from the surface by 54.74°. Since the anisotropic etching is carried out from both

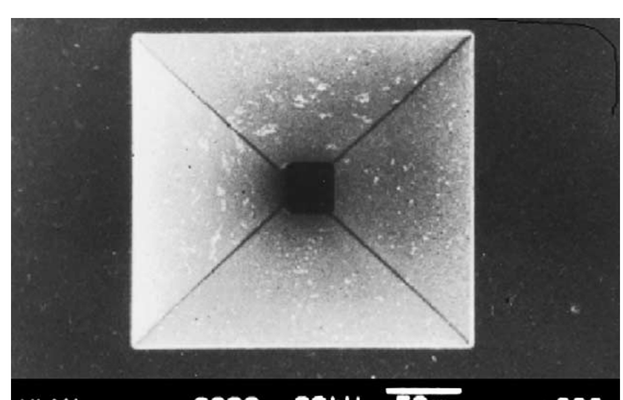

Fig. 1. SEM photograph of the exit nozzle of 30 μm × 30 μm throat size.

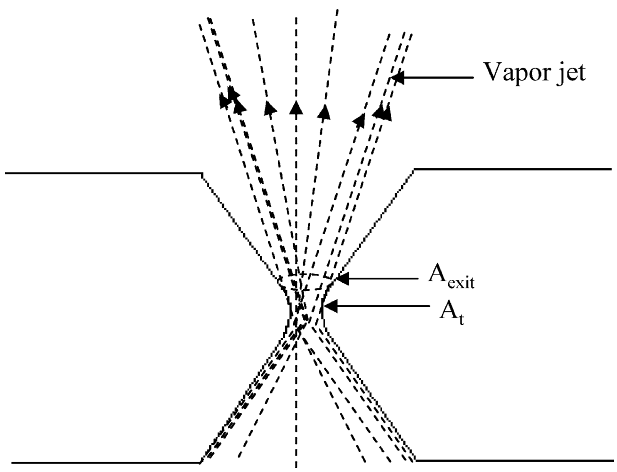

Fig. 2. Cross-sectional view of exit nozzle showing vapor jet flow pattern.

sides, the throat is located at the centre of thickness, i.e., 135 μm below the surface. The side walls of the pyramidally shaped exit nozzle may thus be considered to be too diverging to match with the contour of the emerging vapor jet which is expected to be more directional. In this case, A_{exit} is the cross-sectional area of the vapor jet at the exit side where the vapor stream cuts away from the nozzle wall, as illustrated in Fig. 2. It is not easy to estimate A_{exit}. Hence the thrust cannot be computed in the straightforward manner as discussed above. An iterative scheme will be presented in Section 5 to estimate A_{exit}.

3.2. Heater energy and power

Assuming that there is no loss of heat due to conduction, convection and radiation, the energy supplied by the micro-heater is consumed to heat up the liquid propellant and vaporize it. The total energy (Q_T) involved may be expressed as:

$$Q_T = Q_0 + Q_1 + Q_2 + Q_3 \tag{4}$$

where Q_1 is the energy required to raise the temperature of liquid of mass m to the vaporizing point and Q_2 is the energy for vaporization at this temperature, given by:

$$Q_1 = mc\Delta T \tag{5}$$

$$Q_2 = mL_v \tag{6}$$

where $\Delta T = T_c - T_i$, T_c the vaporization temperature inside the chamber, T_i the initial temperature of the inlet liquid, c the specific heat of the liquid and L_v is the latent heat of vaporization.

Q_0 is the energy consumed to heat up the remaining amount of liquid filling the chamber ($\rho V_c - m$), to temperatures below the vaporization point, where ρ is the density of the liquid and V_c is the chamber volume. It may be noted that the distribution of temperature within the chamber will be highly non-uniform. Hence, it is difficult to express Q_0 analytically.

Q_3 is the amount of heat involved in raising the temperature of vapor above the vaporization point after complete vaporization. Q_3 is given by:

$$Q_3 = \rho V_c C_v (T_c - T_c')$$

where C_v is the specific heat of gas at constant volume and T_c' is the chamber temperature at the point of complete vaporization.

For small values of heater power, the temperature of the water entering the chamber will gradually rise. Thus, Q_0 will be large compared to Q_1 and Q_2, and $Q_3 = 0$ for small heater power levels. However, for intermediate values of heater power, Q_0 will decrease, and both Q_1 and Q_2 will increase as more and more liquid is heated to the vaporization point and vaporized, Q_3 remaining zero. When the entire amount of liquid filling the chamber is heated to vaporization point, $m = \rho V_c$, Q_0 falls to zero and Q_3 will continue to remain zero till the entire mass of the liquid (ρV_c) is completely vaporized. Above this power level, Q_1 and Q_2 will become constant, and Q_3 will start increasing from zero.

If m^* is the mass flow rate of liquid flowing through the chamber (which will also be equal to the mass flow rate of vapor emerging from the nozzle), the fluid filling the chamber will escape through the exit nozzle over a time τ, given by:

$$\tau = \frac{\rho V_c}{m^*} \tag{7}$$

Thus, the average power is given by:

$$W = \frac{Q_T}{\tau} = \frac{m}{\rho V_c} \overset{*}{m} (c\Delta T + L_v) + \frac{m^*}{\rho V_c}(Q_0 + Q_3) \tag{8}$$

The above derivation assumes zero heat loss. In actual practice, however, there will be heat loss due to conduction, convection and radiation. Heat will be conducted away from the top and bottom substrates at the rates:

$$q_{ct} = \frac{K_{Si} A_{avt}}{d_t}(T_c - T_{st}) \tag{9}$$

$$q_{cb} = \frac{K_{Si} A_{avb}}{d_b}(T_c - T_{sb}) \tag{10}$$

where the suffices t and b refer to top and bottom substrates, respectively, A_{av} the average cross-sectional area for conduction, d the thickness, T_s the surface temperature outside the chamber and K_{Si} is the thermal conductivity of silicon. This is based on one-dimensional heat-flow assumption. The temperature of the heater surface and that of the top surface of the cavity are assumed to be T_c which is the chamber temperature during operation.

The chip assembly is mounted on an insulating substrate to prevent heat loss. In the present work, a Pyrex glass was used. Assuming an intimate thermal contact between the bottom chip and the glass substrate, the rate of conduction loss through the glass substrate is:

$$q_{cg} = \frac{K_g A_{sb}}{d_g}(T_{sb} - T_{cg}) \tag{11}$$

where K_g is the thermal conductivity of glass, d_g the thickness of the glass substrate and T_{cg} is the temperature of the bottom surface of the glass substrate. T_{cg} is taken to be 50 °C from practical considerations for the sake of analytical simplicity.

In the steady state,

$$q_{cb} = q_{cg},$$

yielding

$$\frac{T_{sb} - T_{cg}}{T_c - T_{sb}} = \frac{K_{Si}}{K_g}\frac{A_{avb}}{A_{sb}}\frac{d_g}{d_b} \quad (12)$$

This equation is solved to obtain T_{sb}, and hence the heat loss through the bottom substrate is estimated using Eq. (10).

The top substrate is exposed. So, there will be heat loss from the surface through convection and radiation processes, the rates being given by:

$$q_{conv.} = h_c A_{st}(T_{st} - T_i) \quad (13)$$

$$q_{rad.} = \sigma A_{st}(T_{st}^4 - T_0^4) \quad (14)$$

where h_c is the average convective heat transfer coefficient, σ the Stefan–Boltzman constant, A_{st} the effective surface area from which convection and radiation take place and T_0 is the temperature of the surrounding.

In the steady state,

$$q_{ct} = q_{conv.} + q_{rad}, \quad (15)$$

yielding an equation involving T_{st} and other known parameters. Hence, T_{st} may be estimated by solving the equation.

The power generated by the microheater is:

$$H = VI = I^2 R \quad (16)$$

where V is the applied voltage, I the current and R is the resistance. The power delivered by the heater is partly lost by the conduction, convection and radiation processes. The remaining power is utilized to vaporise the liquid. Hence

$$H = W + q_{ct} + q_{cb} \quad (17)$$

where W is given by Eq. (8). As discussed above, q_{ct} and q_{cb} may be evaluated as a function of chamber temperature T_c. Eq. (17) may be rewritten as:

$$\frac{H - q_{ct} - q_{cb}}{m^*} = \frac{m}{\rho V_c}[c\Delta T + L_v] + C_v(T_c - T_c') + \frac{1}{\rho V_c} Q_0 \quad (18)$$

In the left hand side of the Eq. (18), H and m^* are controllable quantities, and q_{ct}, q_{cb} depend on the chamber temperature T_c. In the RHS, the unknown quantities are m and Q_0. T_c' is the vaporization temperature when the liquid entering the chamber is completely vaporized. It may be estimated by a method to be presented in Section 5. As mentioned earlier, it is difficult to estimate Q_0. But Q_0 becomes zero when the entire amount of liquid filling the chamber (ρV_c) is raised to the vaporization temperature. So, under usual circumstances, $Q_0 = 0$. As more and more liquid is vaporized, $\frac{m}{\rho V_c}$ increases but Q_3 and hence the second term in the RHS remain zero. At an input power level, when complete vaporization takes place, $\frac{m}{\rho V_c} = 1$ and ΔT gets saturated at $\Delta T_{max} = T_c' - T_i$. Above this power level, Q_3 and hence the second term in RHS increase from zero. Under this situation, the Eq. (18) is well defined with no unknowns. It may be solved to obtain T_c. Once T_c is obtained, the chamber pressure P_c may be estimated by applying the principle of thermodynamics. Hence, the thrust may be estimated as a function of heater power and liquid flow rate using the thrust equation described in Section 3.1.

The model is, however, quite approximate as it is based on many simplifying assumptions.

4. Experiments

4.1. Fabrication

The VLM fabricated in the laboratory [7] consists two silicon chips bonded together. (1 0 0) n-Si wafer of resistivity 4–6 Ω cm was processed using the bulk micromachining technique based on wet anisotropic etching. The top wafer containing the inlet and exit nozzles and the upper part of the vaporizing chamber was processed by anisotropic etching in KOH solution. It involved two photomasks and double-side alignment. The bottom wafer contains p-diffused meander-line resistor, the lower part of cavity and the V-groove microchannel for liquid inflow. The details of the fabrication process are given in ref. [7]. A schematic of the test device and a photograph of the same are shown in Fig. 3 (a and b), respectively.

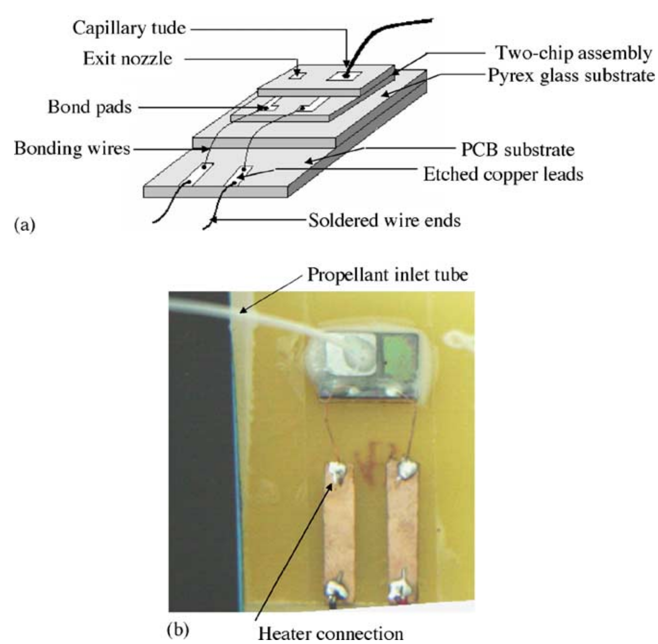

Fig. 3. (a) Schematic view of test device and (b) microphotograph of the VLM test device.

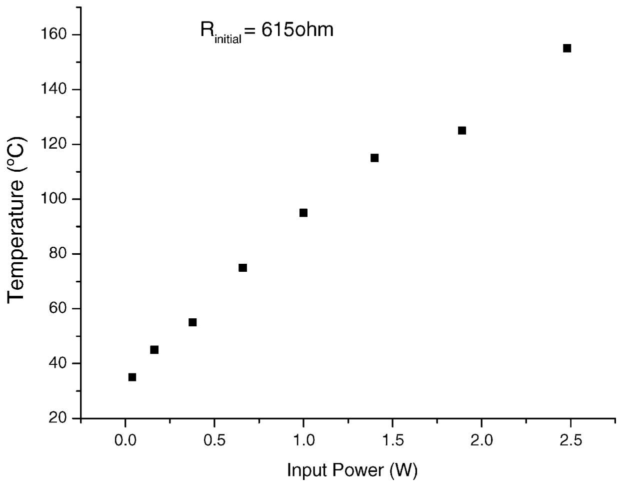

Fig. 4. Measured temperature vs. applied heater power in dry condition.

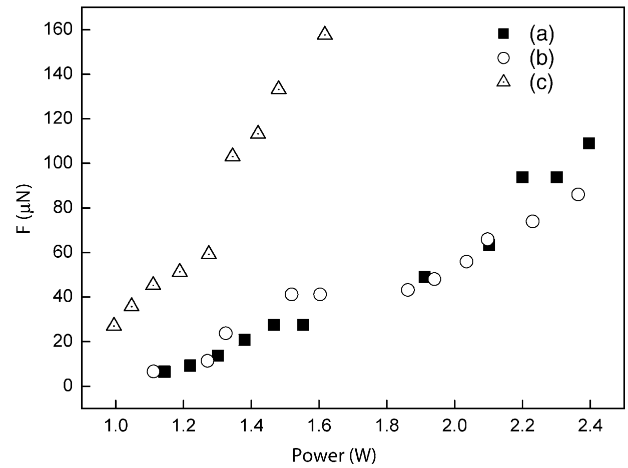

Fig. 5. Measured thrust vs. heater power plots for (a) flow rate 1 μl/s, exit nozzle throat 30 μm × 30 μm, (b) flow rate 1.6 μl/s, exit nozzle 30 μm × 30 μm and (c) flow rate 0.8 μl/s, exit nozzle 50 μm × 50 μm.

4.2. Testing and measurements

A sensitive cantilever made of a thin copper foil strip was used for the purpose of testing and measurements as described in [7]. The over all dimensions of the copper foil cantilever were 45 mm length, 5 mm width and 0.05 mm thickness. The lamp-and-scale arrangement was used for the testing purpose. Using this method the small deflection of the cantilever was effectively magnified by several orders of magnitude. The thrust or force acting on the cantilever is given by:

$$F = K\left[\tan^{-1}\left(\frac{h_1}{L}\right) - \tan^{-1}\left(\frac{h_2}{L}\right)\right] \quad (19)$$

where K is a constant dependent on the geometry of the cantilever and the elastic constant of the cantilever material, L the horizontal distance between the cantilever and the screen; h_1 and h_2 are the height of the light spot on the screen from the basal plane (horizontal) of the cantilever before and after the application of the thrust with reference to Fig. 4 in ref. [7]. For the particular setup, $K = 71.0 \times 10^{-6}$ N.

For the sake of experimental simplicity, deionised water was used as the liquid propellant, which has also been reported by previous workers [3,4], although water suffers from the drawback due to its large heat of vaporization. During the measurement the gap between the bottom surface of the cantilever end and the top surface of the VLM was maintained at 4 mm, with the presumption that the thrust calculated by using Eq. (19) is the actual thrust generated by the VLM. A gravity controlled arrangement was made for adjusting the flow rate of water. The heater power was controlled by adjusting the voltage of a DC power supply connected to the heater terminals. The heater surface temperature was calibrated against input electrical power, under dry condition without any water flow, using a thin copper–constantan thermocouple whose tip was kept in contact with the oxidized silicon surface close to the diffused heater. Fig. 4 shows the temperature versus power plot in dry condition. For the particular device, it was found that a minimum heater power of 1 W was required to raise the temperature to 100 °C without any water flow. The heater power was varied from 1 to 2.5 W in the present study. Larger heater power was not used to avoid overheating that led to unpredictable failures.

Fig. 5 shows the variation of the measured thrust with the heater power for three cases: (a) flow rate 1 μl/s, exit nozzle throat 30 μm × 30 μm, (b) flow rate 1.6 μl/s, exit nozzle 30 μm × 30 μm and (c) flow rate 0.8 μl/s, exit nozzle 50 μm × 50 μm. For the 30 μm × 30 μm nozzle, the curves for 1 μl/s and 1.6 μl/s flow rates cross each other at about 1.8 W. At power levels below the crossover points, the thrust was found to be higher for the lower flow rate, and vice versa. For the broader exit nozzle (50 μm × 50 μm), higher thrusts were generated at lower power levels and at a lower flow rate.

5. Discussion

It has already been pointed out in Section 3 that the theoretical model proposed for the VLM is purely analytical and too approximate to be used for exact quantitative verification. However, the physical principles on which the model is based are valid. In order to interpret the experimental results, a back-to-front approach will be followed.

We start with experimental curves of thrust versus power for $\frac{m^*}{\rho} = 1$ μl/s (water) and $A_t = 30$ μm × 30 μm, as shown in Fig. 5. As discussed in Section 3.1, in the case of exit nozzle formed by wet anisotropic etching, A_{exit} has to be found by an iterative approach. It will be convenient if the thrust is expressed as:

$$F = X_1 + X_2 \quad (20)$$

where

$$X_1 = \overset{*}{m}\, V_{exit} \quad (21)$$

and

$$X_2 = (P_{\text{exit}} - P_{\text{atm}})A_{\text{exit}} \tag{22}$$

V_{exit} and $\left(\frac{A_{\text{exit}}}{A_t}\right)$ are given by Eqs. (2) and (3).

It is to be noted that T_c and P_c are interrelated thermodynamically. P_c is taken to be the saturation vapor pressure under the assumption that the nozzle throat area is small compared with the internal surface area of the vaporizing chamber. Since water is used as the propellant in the present work, the steam table [8] may be used for correlating T_c and P_c. The linear interpolation is used to get the values of P_c at temperatures in between the tabulated values.

Starting from the values of F_{measured} corresponding to the measured values of $\frac{m^*}{\rho}$, A_t and H, the back-to-front iterative approach to extract the values of A_{exit}, T_c, P_c and P_{exit} is as follows:

1. (a) Assume $X_2 = 0$. Hence, $F = X_1 = m^* V_{\text{exit}}$. Starting with $F = F_{\text{measured}}$, calculate $V_{\text{exit}} = V_{\text{exit}}^{(1)}$.
   (b) Assume T_c slightly above 100 °C (373 °K), say $T_c{}^{(1)}$. Hence calculate $\left(\frac{P_{\text{exit}}}{P_c}\right)_1$ for $V_{\text{exit}} = V_{\text{exit}}^{(1)}$ using Eq. (2). Find $P_c = P_c{}^{(1)}$ from steam table and calculate $\left(\frac{A_{\text{exit}}}{A_t}\right)_1$ and hence $A_{\text{exit}}^{(1)}$ using Eq. (3), since A_t is known.
   (c) Compute $X_2^{(1)} = (P_{\text{exit}}^{(1)} - P_{\text{atm}})A_{\text{exit}}^{(1)}$. In this computation, the barometric pressure (P_{atm}) is taken to be 0.101415 MPa valid for Kharagpur where the experimental testing was carried out.
   (d) Compute $F^{(1)} = X_1 + X_2{}^{(1)} = m^* V_{\text{exit}}^{(1)} + X_2{}^{(1)}$. Note that $F^{(1)} > F_{\text{measured}}$.
2. (a) In the next step, $F_{\text{measured}} - X_2^{(1)} = m^* V_{\text{exit}}^{(2)}$. Hence compute $V_{\text{exit}}^{(2)}$.
   (b) Take $T_c^{(2)} = T_c^{(1)} - \delta T$. Find $P_c^{(2)}$ from steam table. Hence, calculate $\left(\frac{P_{\text{exit}}}{P_c}\right)_2$ and $P_{\text{exit}}^{(2)}$. Then compute $\left(\frac{A_{\text{exit}}}{A_t}\right)_2$ and $A_{\text{exit}}^{(2)}$.
   (c) Compute $X_2^{(2)} = \left(P_{\text{exit}}^{(2)} - P_{\text{atm}}\right) A_{\text{exit}}^{(2)}$.
   (d) Compute $F^{(2)} = X_1^{(1)} + X_2^{(2)} = m^* V_{\text{exit}}^{(2)} + X_2^{(2)}$. $F^{(2)}$ is expected to be less than $F^{(1)}$.
3. (a) Next, $F_{\text{measured}} - X_2^{(2)} = m^* V_{\text{exit}}^{(3)}$. Hence compute $V_{\text{exit}}^{(3)}$.
   (b) Take $T_c^{(3)} = T_c^{(2)} - \delta T'$. Find $P_c^{(3)}$ from steam table. Hence calculate $\left(\frac{P_{\text{exit}}}{P_c}\right)_3$ and $P_{\text{exit}}^{(3)}$. Then compute $\left(\frac{A_{\text{exit}}}{A_t}\right)_3$ and $A_{\text{exit}}^{(3)}$.
   (c) Continue.

This computing process continuous till F^n converges to F_{measured}. Hence the corresponding values of T_c, P_c, P_{exit} and A_{exit} are extracted. This iterative computation is repeated for all measured values of thrust.

In Fig. 6, the extracted values of T_c are plotted against the measured values of thrust for the 30 μm × 30 μm nozzle, for 1 and 1.6 μl/s water flow rates. The figure also includes the variation of A_{exit} with T_c for 1.6 μl/s flow rate of water, as obtained from the above computations. The variation of T_c (extracted) with the heater power is shown in Fig. 7 for 30 μm × 30 μm nozzle and 1 μl/s water flow rate. The results may be discussed from the physical point of view as follows:

For the present device, a minimum heater power of 1 W was required to raise the heater temperature to 100 °C without any water flow. As water is introduced into the chamber, it will be heated to pre-boiling condition causing high evaporation. Hence a thrust will be created due to emerging vapor jet. The evaporation rate will increase with temperature of water with increasing heater power, leading to a rise of vapor pressure and thrust. However, as vaporization or boiling starts, the temperature cannot change appreciably, the applied heat being consumed for supplying the latent heat of vaporization. The vaporization temperature will depend on the internal vapor pressure which may be assumed to be the saturation vapor pressure, the exit nozzle area being small compared with the

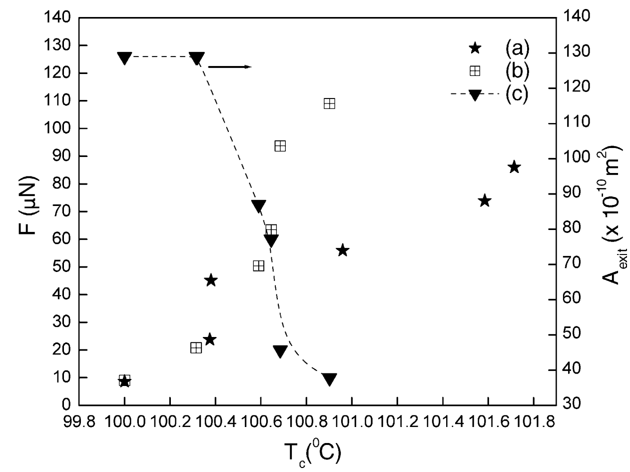

Fig. 6. Extracted values of T_c vs. measured value of thrust for exit nozzle throat 30 μm × 30 μm with flow rate (a) 1 μl/s and (b) 1.6 μl/s; and variation of A_{exit} with extracted value of T_c for flow rate1.6 μl/s and exit nozzle throat 30 μm × 30 μm (c).

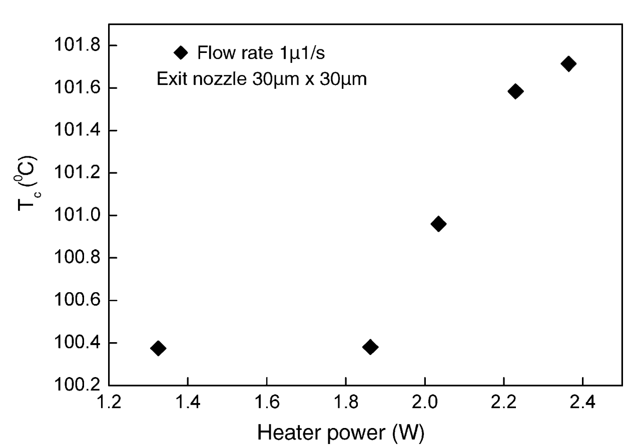

Fig. 7. Chamber temperature as a function of applied heater power for flow rate1 μl/s.

internal surface area of the chamber. When the liquid filling the chamber is fully vaporized, the phase transformation is completed. Beyond this point, the vapor temperature and pressure should increase with the further increase of heater power. Hence, the thrust will also increase. The curves in Figs. 6 and 7 clearly demonstrate the above physical phenomena. The phase transformation takes place at 100.4 and 100.65 °C for the water flow rates 1 and 1.6 μl/s, respectively. During the phase transformation, the thrust increases from 20 to 45 μN for 1 μl/s and from 50 to 90 μN for 1.6 μl/s. For the higher flow rate of water, it requires more heating power to initiate boiling. But a stronger vapor pressure will be created, leading to an increased thrust. The higher vapor pressure will result in a higher vaporization temperature.

The variation of A_{exit} with T_{c} (Fig. 6) is quite revealing. Upto the vaporization temperature A_{exit} is relatively large and constant. Above this temperature, A_{exit} decreases continuously. Before the onset of phase transformation, vapor pressure will be low and hence the vapor jet will be less energetic and more diffusive, resulting in higher A_{exit}. But during and after phase transformation, the vapor pressure and temperature go on increasing, initially at a slower rate, and, thereafter, at a higher rate. Hence, the vapor will emerge with a higher kinetic energy and hence with a higher exit velocity. The energetic jet will be more directional and hence A_{exit} will decrease.

Finally, an attempt is made to correlate T_{c} with the heater power (H) for the curve shown in Fig. 7 using Eq. (18). During the phase transformation from the onset of vaporization to the point of complete vaporization, $Q_0 = 0$ and $Q_3 = 0$. Hence the second and third terms in the RHS of Eq. (18) will disappear. In this case, the fraction of liquid vaporized $\frac{m}{\rho V_{\mathrm{c}}}$ may be computed as a function of the heater power (H). The two will be linearly related, since T_{c} and P_{c} will remain nearly constant. q_{ct} and q_{cb} may be estimated using the simple model presented in Section 3.2. The rate of increase of $\frac{m}{\rho V_{\mathrm{c}}}$ with heater power during phase transformation is thus found to be about 0.5 per W for a typical test device. It means that a heater power of about 2 W is required for the complete vaporization of water filling the chamber. This is, however very approximate, since the simple models for the estimation of q_{cb} and q_{ct}, as described in Section 3.2, are not sufficiently accurate.

6. Conclusions

A simple theoretical model of a MEMS VLM has been developed. The back thrust generated by the emerging vapor jet through a nozzle may be estimated using the thrust equation available in the literature. It, however, requires the values of chamber temperature and pressure as a function of heater power, the mass flow rate and the exit pressure related to the nozzle expansion ratio. An approximate one-dimensional thermal model has been proposed to estimate chamber temperature in terms of heater power. The chamber temperature and pressure are interrelated thermodynamically. Using water as the propellant, the steam table is used for this purpose assuming that the exit nozzle throat area is very small compared with the internal surface area of the chamber. The anisotropically etched exit nozzle being too divergent to match the contour of the emerging vapor jet, it was difficult to estimate directly the cross-sectional area of the vapor jet at the actual exit point. Hence, an iterative back-to-front approach was adopted to estimate the effective exit area from the measured value of thrust of a VLM fabricated in the authors' laboratory. It was, thus, possible to extract the values of the chamber temperature and pressure and the effective exit area for each value of thrust measured as a function of the heater power. The results have been discussed and interpreted physically. It may thus be concluded that the simple model developed in this paper cannot be used for a precise quantitative simulation of a MEMS VLM, but it will serve as an effective tool for the understanding of the physical processes and for a rough estimation of useful parameters required during the development of a VLM based on MEMS technology.

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Biographies

D.K. Maurya was born in India in 1975. He received MSc in electronics from DDU Gorakhpur University in 1998, and MS in electronics and

electrical communication engineering from Indian Institute of Technology, Kharagpur in 2004. From March 2000 to January 2001, he worked for Continental Device India Limited, New Delhi, as Technical Officer, where he was involved in the processing of discreet semiconductor devices. Since January 2001, he has been working as Scientific Officer in the Advanced Technology Centre of IIT, Kharagpur. He is actively involved in the process development, fabrication and characterization of MEMS devices. His research interests include MEMS and Microelectronics, microsensors and microactuators, microfabrication and micromachining technologies.

S. Das received his MSc in physics from the IIT, Kharagpur, in 1988. He joined at the Microelectronics Centre, Department of Electronics and ECE, IIT Kharagpur, in 1989 as a CSIR Research Fellow. He received his PhD in 1996 on large bias conduction in polysilicon, high current pulse trimming of polyresistors and transient studies of polysilicon fuses. He worked as Scientific Officer in Microelectronics Centre in E&ECE Department during the period 1996–2000. Since 2000, he has been working as Sr. Scientific Officer in Advanced Technology Centre, IIT, Kharagpur. His research areas include MEMS and micromachining technology, inertial sensors and actuators, thin film processing for VLSI, bipolar devices, etc. He is actively involved in the development of silicon and quartz micromachining technology and microsensors, and fabrication and characterization of bipolar devices and ICs. He has also been involved in experimental research on ion beam processing for VLSI like dual ion beam deposition of oxynitride film of variable composition, chemically assisted ion beam etching and polysilicon film deposition and characterization.

S.K. Lahiri was born at Kolkata in 1943. He received M.Tech. and PhD in radio physics and electronics from the University of Calcutta in 1966 and 1971, respectively. He joined Indian Institute of Technology, Kharagpur as a faculty member in November 1971 and has been continuing as the professor of Department of Electronics and Electrical Communication Engineering in IIT Kharagpur since July 1985. His areas of research include microelectronics/VLSI, MEMS and integrated optics. He has authored about 150 research papers in journals and proceedings of conferences, and supervised 12 PhD theses and more than 40 M.Tech/MS theses. He served as the Dean-Sponsored Research & Industrial Consultancy (1997–2000) and the deputy director of IIT Kharagpur (2002–2005) and also chaired Advanced Technology Centre, Advanced VLSI Design Laboratory and Space Technology Cell in the Institute. Prof. Lahiri is a Fellow of Indian National Academy of Engineering.